

Question Paper

Exam Date & Time: 28-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Basic Electrical Technology [ELE-1071]

Marks: 50

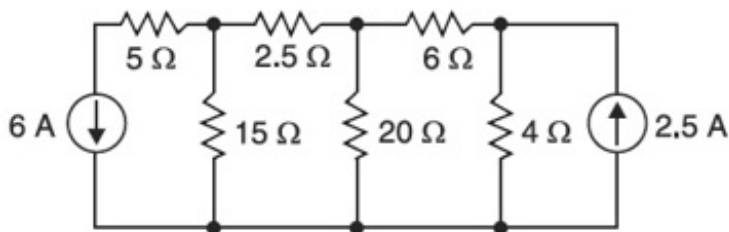
Duration: 180 mins.

PART-A

Answer all the questions.

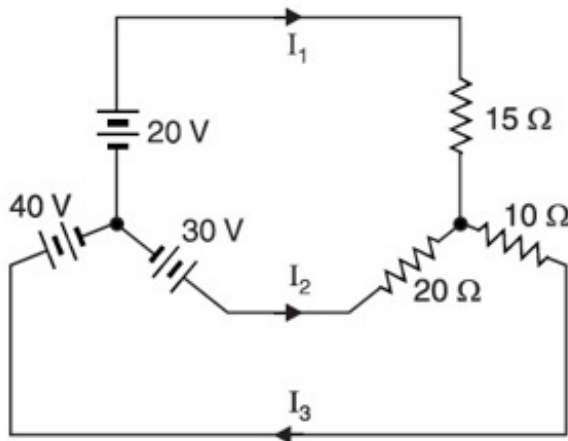
- 1) Find the current flowing through 5Ω and 4Ω resistor using nodal analysis. (5)

A)



Course Outcome: "CO1" / Cognitive Level: "Application"

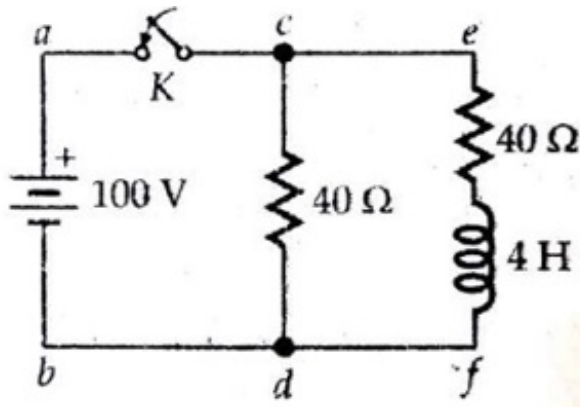
- B) Using superposition theorem, find the current in each branch of the network shown in figure below. (5)



Course Outcome: "CO1" / Cognitive Level: "Application"

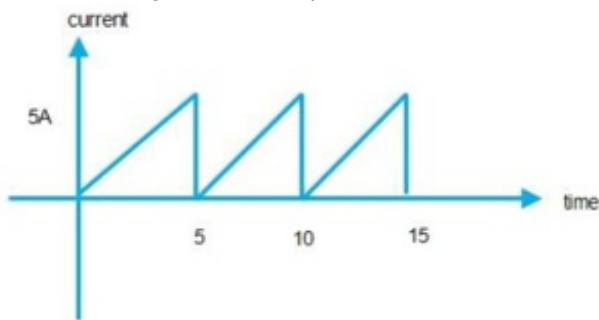
- 2) In the figure shown below after the steady state condition is reached, at time $t=0$ the switch K is suddenly opened. Find the value of the current through the inductor at time $t = 0.5$ seconds. (4)

A)



Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Find the average, rms value, peak and form factor for the circuit shown in figure below. (3)



Course Outcome: "CO2" / Cognitive Level: "Analysis"

- C) A resistance of 7 is connected in series with a pure inductance of 31.8 mH and the circuit is connected to a 100 V, 50 Hz sinusoidal supply. Calculate (3)
- circuit current
 - Phase angle
 - Power factor
 - Power.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Two circuits, the impedances of which are given by $Z_1 = (15 + j12)$ ohms and $Z_2 = (8 + j5)$ ohms are connected in parallel. If the potential difference across one of the impedances is $(250 + j0)$ V, calculate (5)
- A)
- total current and branch currents
 - total power and power consumed in each branch and
 - overall power-factor

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Two-wattmeter method is used to determine the input power to a three-phase motor. The readings of the meters are 5.2 kW and -1.7 kW, and the line voltage is 415 V. Calculate (5)
- the total power,
 - the power factor, and
 - the line current.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) A balanced star connected load of $(8+j6)$ Ω per phase is connected to a 3 ϕ , 230 Volts, 50 Hz, AC supply. Find (6)
- A)
- Line current
 - Power factor
 - Power

- iv) Total reactive Power
v) Total Volt-Amp

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) An iron ring of cross-sectional area 6 cm^2 is wound with a wire of 100 turns and has a saw cut of 2 mm. Calculate the magnetising current required to produce a flux of 0.1 mWb if mean length of magnetic path is 30 cm and relative permeability of iron is 470. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) A cast steel electromagnet has an air gap length of 3 mm and an iron path of length 40 cm. Find the number of ampere-turns necessary to produce a flux density of 0.7 Wb/m^2 in the gap. Neglect leakage and fringing. Assume ampere-turns required for air gap to be 70% of the total ampere-turns. (3)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Explain the construction of single-phase induction motor. (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- C) Explain the working of single-phase digital energy meter with a neat diagram. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ELE-1071:Basic Electrical Technology

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	11
3	3	CO 3	11
4	4	CO 4	7
5	5	CO 5	7
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ELE-1071:Basic Electrical Technology

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	7(14%)
3	Application	5	23(46%)
4	Analysis	5	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	50 (100%)